

Spark Structured Streaming - Join Rules (Quick Reference)

This document is a concise, production-ready cheat sheet for join rules in Spark Structured Streaming.

Default Rule

Default join type in Spark is INNER JOIN.

1. Static JOIN Static (Batch)

All join types allowed: inner, left, right, full.

No watermark required.

No time condition required.

2. Stream JOIN Static (or Static JOIN Stream)

All join types allowed: inner, left, right, full.

No watermark required.

Commonly used for enrichment.

```
orders_stream.join(users_static, "user_id", "left")
```

3. Stream JOIN Stream - INNER JOIN

Allowed without watermark.

No time condition required.

Output mode: append.

```
orders.join(status, "order_id", "inner")
```

4. Stream JOIN Stream - LEFT / RIGHT OUTER JOIN

Restricted. All conditions below are mandatory:

Event-time column required.

Watermark required.

Time-bounded join condition required.

Output mode: append.

```
orders.withWatermark("order_time", "5 minutes") .join(  
  status.withWatermark("status_time", "5 minutes"), orders.order_id = status.order_id  
  AND status.status_time BETWEEN orders.order_time AND orders.order_time + interval 5  
  minutes, "leftOuter" )
```

5. Stream JOIN Stream - FULL OUTER JOIN

Not supported in Structured Streaming.

Reason: infinite state, no safe eviction.

6. Quick Decision Table

Join scenario	Allowed	Watermark needed
Static JOIN Static	Yes	No
Stream JOIN Static	Yes	No
Stream JOIN Stream (inner)	Yes	No

Stream JOIN Stream (left/right)	Conditional	Yes
Stream JOIN Stream (full)	No	-

7. One Rule to Remember

If Spark cannot prove that join state is bounded in time, the join is rejected.

8. Mental Shortcuts

Inner join = easy.

Stream-Static join = safe.

Outer stream joins = time-bounded only.

No watermark = Spark refuses.